

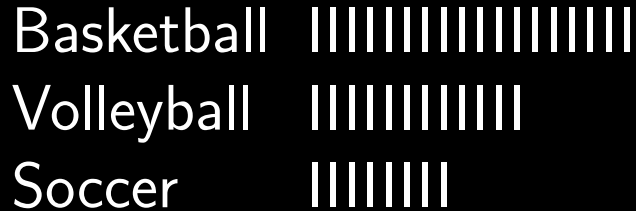
Graphical Presentation of Statistical Data

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Graphs are visual representations of organized data.

Bar Graph

A **Bar Graph** is used to compare quantities among different categories.



Taller bars mean bigger quantities.

Line Graph

A **Line Graph** is used to show changes or trends over time.

Jan → Feb → Mar → Apr

2 → 3 → 5 → 6

The line moves upward, showing an increasing trend.

Pie Graph

A **Pie Graph** (or circle graph) shows how each category represents a part of the whole.

If milk = $\frac{1}{2}$ and juice = $\frac{1}{4}$, together they make $\frac{3}{4}$ of the circle.

The whole circle always equals 1 (or 100%).

Stem-and-Leaf Plot

A **Stem-and-Leaf Plot** organizes numerical data while preserving the original values.

Stem	Leaf
2	1 4 6
3	1 3 4

Stem 2, leaf 1 means the value 21.

Which Graph Should I Use?

Purpose	Best Graph
Compare categories	Bar Graph
Show change over time	Line Graph
Show parts of a whole	Pie Graph
Keep exact values, organized	Stem-and-Leaf

Examples

Example 1

Favorite fruits: Apple = 10, Banana = 7,
Mango = 5

Since we are **comparing categories**, the best choice is a **Bar Graph**.

Example 2

Daily temperature for a week:

28, 29, 27, 30, 31, 29, 28 (in °C)

Since we are showing **change over time**, the best choice is a **Line Graph**.

Example 3

A class budget shows: Snacks = $\frac{1}{2}$, Supplies = $\frac{3}{10}$, Savings = rest.

Since we are showing **parts of a whole**, the best choice is a **Pie Graph**.

$$\text{Savings} = 1 - \frac{1}{2} - \frac{3}{10} = \frac{1}{5}$$

Example 4

Quiz scores: 32, 35, 38, 41, 44, 47

Stem	Leaf
3	2 5 8
4	1 4 7

Since we want to **keep the exact scores organized**, a **Stem-and-Leaf Plot** works best.

Practice Exercises

Practice 1

A survey of favorite sports produced the following data: Basketball = 18, Volleyball = 12, Soccer = 8, Badminton = 6

Which graph is most suitable to compare these categories?

Practice 2

The number of books read by Ana during the first six months is shown: Jan = 2, Feb = 3, Mar = 5, Apr = 6, May = 4, Jun = 7

Which graph best represents this information?

Practice 3

A pie graph shows that $\frac{1}{2}$ of the class prefers milk, $\frac{1}{4}$ prefers juice, and the rest prefer water. What fraction of the class prefers water?

Practice 4

Choose the most appropriate graph for each situation.

1. Comparing sales of four stores
2. Showing rainfall over twelve months
3. Displaying the parts of a school's budget
4. Organizing students' quiz scores

Activities

Activity 1

A survey of favorite pets produced the following data: Dogs = 20, Cats = 15, Birds = 7, Fish = 5

Which graph is most suitable to compare these categories?

Activity 2

The number of plants grown by Ben during the first six months is shown: Jan = 4, Feb = 5, Mar = 6, Apr = 7, May = 8, Jun = 10

Which graph best represents this information?

Activity 3

A pie graph shows that $\frac{3}{5}$ of the class prefers reading, $\frac{1}{5}$ prefers drawing, and the rest prefer sports. What fraction of the class prefers sports?

Activity 4

Choose the most appropriate graph for each situation.

1. Comparing the number of vehicles in different towns
2. Showing electricity consumption for twelve months
3. Displaying the distribution of household expenses
4. Organizing examination scores

Thank you!